

Computational Neuroscience Through a Kuhnian Lens

Executive summary

Computational neuroscience is best described as a *multi-paradigm field* rather than a mature Kuhnian “normal science” organized around a single dominant paradigm. This pluralism is partly structural: the field spans multiple explanatory levels (from ion channels to behavior) and multiple success criteria (mechanistic realism, normative optimality, predictive accuracy, engineering performance). These differences make Kuhnian “incommensurability” risks unusually salient: paradigms often talk past each other because they aim at different targets (e.g., *mechanism vs function*, *neural prediction vs behavioral prediction*). ¹

Across the paradigms requested (single-neuron biophysics; rate and spiking networks; Bayesian/probabilistic coding; predictive coding; deep learning; reinforcement learning; dynamical systems/attractors; connectomics), each has clear “normal science” puzzle sets and major successes anchored in canonical exemplars (e.g., Hodgkin–Huxley conductance modeling; Wilson–Cowan population dynamics; Hopfield attractors; Bayesian cue integration; predictive coding hierarchies; task-trained deep nets predicting ventral-stream responses; dopamine-linked RL; low-dimensional neural dynamics; large-scale connectome reconstructions). ²

However, each paradigm also faces concrete, well-documented anomalies or unresolved tensions. Many anomalies are *persistent but local*: they challenge a paradigm’s scope conditions (when it works) more than its core existence. Examples include (i) parameter degeneracy and identifiability limits in detailed conductance/circuit models; (ii) empirical evidence that “expectation suppression” and repetition effects can dissociate, complicating strong predictive-coding readings; (iii) systematic mismatches between standard convolutional neural networks and human vision/robustness; (iv) dopamine signals reflecting movement/salience dimensions beyond a simple scalar reward prediction error; and (v) the limited ability of “more data” approaches (including connectomics-style mapping plus standard analysis pipelines) to automatically yield mechanistic understanding. ³

On Kuhn’s criteria for crisis—persistent anomalies, loss of confidence, and proliferation of alternatives—computational neuroscience shows *partial crisis signals* in a few contested interfaces (notably: the “Bayesian optimality” program, predictive processing claims as a unified cortical theory, and deep neural networks as explanatory brain models), but **not** a field-wide crisis that would plausibly culminate in a single paradigm replacing another. Instead, the dominant pattern since roughly the 2010s is *proliferation plus attempted synthesis* (benchmarking platforms, hybrid models, and calls to integrate cognitive modeling, neuroscience, and AI), which is consistent with an expanding, interdisciplinary research program rather than a Kuhnian revolution. ⁴

Assumptions adopted (as requested): the target audience is an informed academic reader; the depth is intended as a comprehensive, analytic review (bounded by what can be supported with primary sources and major reviews). ⁵

Kuhnian framework applied to computational neuroscience

Kuhn's core units for analysis are *paradigms* (shared exemplars, methods, standards, and problem-sets), *normal science* (puzzle-solving within those standards), *anomalies* (persistent puzzle failures), and *crisis/revolution* (loss of confidence and eventual replacement by a rival paradigm). Kuhn also emphasizes that anomalies alone rarely topple a paradigm unless credible alternatives exist, and that paradigm competition involves shifts in what counts as a legitimate problem and solution. ⁶

Two adaptations are necessary for computational neuroscience:

First, computational neuroscience is explicitly multi-level. **David Marr** ⁷ 's three levels (computational theory, algorithm/representation, implementation) provide a widely used lens for why different "paradigms" may not be direct competitors: they can address different levels of the same phenomenon rather than offering mutually exclusive accounts. ⁸

Second, computational neuroscience inherits diverse epistemic standards from neuroscience, statistics, physics, and AI. For example, some paradigms value *biophysical realism* and parameter interpretability; others prioritize *predictive accuracy* on held-out neural data; others prioritize *normative optimality* or *behavioral fit*. Across these standards, the same dataset can generate "success" in one paradigm and "anomaly" in another, a situation Kuhn would treat as fertile ground for incommensurable disputes. ⁹

Operationalization used in this report:

- A "paradigm" is treated as a Kuhnian family of exemplars + methods + standards (e.g., the Hodgkin-Huxley tradition; the Wilson-Cowan tradition; predictive processing under hierarchical generative models). ¹⁰
- An "anomaly" is a persistent empirical or methodological tension that prompts (i) ad hoc patches, (ii) explicit boundary-setting ("works only for X"), or (iii) the rise of alternative frameworks. ¹¹
- "Crisis signals" are evaluated by (a) persistence/centrality of anomalies, (b) explicit loss of confidence in major venues (target articles, high-impact reviews, methodological critiques), and (c) proliferation of rival frameworks or benchmarking regimes. ¹²

Paradigm map and cross-paradigm comparison

Computational neuroscience can be organized into interacting paradigm clusters rather than a linear sequence. The map below emphasizes (i) explanatory level, (ii) typical data targets, and (iii) competition vs complementarity.

```
flowchart TB
    subgraph Mechanistic_Implementation["Mechanistic implementation"]
        SNB["Single-neuron biophysics\n(conductance/compartement models)"]
        SNN["Spiking networks\n(point neurons, STDP, spike-train models)"]
        RBN["Rate-based networks\n(mean-field/population ODEs)"]
    end
    end

    subgraph Normative_Inference["Normative & inference frameworks"]
```

```

    BB["Bayesian brain / probabilistic coding\n(ideal observers, PPCs)"]
    PC["Predictive coding / predictive processing\n(hierarchical generative
models)"]
    RL["Reinforcement learning models\n(TD errors, policies, value)"]
end

subgraph Systems_Modeling["Systems-level modeling styles"]
    DS["Dynamical systems & attractors\n(manifolds, fixed points, RNN
dynamics)"]
    DL["Deep learning / neural networks\n(task-optimized models, benchmarks)"]
end

subgraph Structural_Mapping["Structural mapping"]
    CX["Connectomics\n(EM reconstructions, diffusion MRI, graphs)"]
end

SNB --> SNN
SNN <--> RBN
BB <--> PC
RL <--> BB
DS <--> RBN
DS <--> SNN
DL <--> DS
DL <--> BB
CX --> DS
CX --> SNN
CX --> RBN

```

This structure reflects well-established exemplars and reviews: conductance modeling anchored by Hodgkin–Huxley and simulation environments such as **NEURON** ¹³; population dynamics rooted in Wilson–Cowan; attractors rooted in Hopfield; Bayesian/probabilistic coding anchored by Bayesian-brain and probabilistic population code formalisms; predictive processing anchored in hierarchical predictive coding and free-energy formulations; deep learning anchored in task-trained networks predicting ventral stream responses; RL anchored in temporal-difference learning texts and dopamine physiology; connectomics anchored in “connectome” framing and large-scale reconstructions. ¹⁴

Comparative table of paradigms

The table below compresses “disciplinary matrix” elements (assumptions, methods, exemplars, anomalies) in a Kuhnian spirit. Representative papers are primary where possible; reviews are used where they frame debates or synthesize evidence.

Paradigm	Core assumptions	Typical methods	Canonical successes	Concrete anomalies or unresolved tensions	Key proponents	Representative source
Single-neuron biophysics	Spikes arise from biophysical conductances; geometry + channels matter; mechanistic parameterization can explain neural responses	Hodgkin–Huxley-type ODEs, compartmental modeling, parameter fitting; simulation tooling (e.g., NEURON)	Quantitative action potential model; detailed neuron/network simulations; mechanistic linking of channels to excitability	Degeneracy/identifiability: many parameter sets produce similar outputs (channel variability across animals; “similar activity from disparate parameters”; “sloppy” sensitivities); scaling detailed models from sparse data (digital microcircuit reconstructions)	Alan Hodgkin ¹⁵ , Andrew Huxley ¹⁶ ; Eve Marder ¹⁷ ; Henry Markram ¹⁸	¹⁹
Rate-based networks	Population firing rates are sufficient state variables for many computations; spikes can be coarse-grained	Nonlinear ODEs/PDEs, mean-field, stability/phase-plane analyses	Understanding E–I interactions, oscillations, gain control; tractable theory for circuits	Spike-timing and correlation sensitivity can matter: reliable millisecond timing under fluctuating inputs; population correlations carry information beyond pairwise effects; irregular spiking challenges naive integrator assumptions	Hugh Wilson ²⁰ ; Jack Cowan ²¹	²²

Paradigm	Core assumptions	Typical methods	Canonical successes	Concrete anomalies or unresolved tensions	Key proponents	Representative source
Spiking networks	Spikes are meaningful events; timing, synchrony, and plasticity rules can implement computation	Integrate-and-fire and point-process models; spike-train statistics; STDP; multi-neuron decoding	Capturing precise timing and variability; linking synaptic rules to learning; population coding models with spikes	Coding disputes (what counts as “code” vs dynamics); competing empirical signatures (reliability vs variability); learning in spiking nets often requires nontrivial training assumptions (ongoing debate about biological plausibility)	Wulfram Gerstner ²³ ; Eugene Izhikevich ²⁴ ; Romain Brette ²⁵	²⁶
Bayesian brain / probabilistic coding	Perception/ action approximate Bayesian inference; uncertainty represented and combined; behavior may be near-optimal under constraints	Ideal observer models; Bayesian parameter estimation; probabilistic population codes; psychophysics fits	Bayesian cue integration in sensorimotor learning; formal probabilistic population codes framework	Suboptimality and flexibility critiques: substantial suboptimalities documented; concern that priors/likelihoods can be adjusted post hoc (“just-so stories”); early claims highlighted lack of neurophysiological tests of uncertainty coding	Alexandre Pouget ²⁷ ; Daniel Wolpert ²⁸ ; Konrad Körding ²⁹ ; Wei Ji Ma ³⁰	³¹

Paradigm	Core assumptions	Typical methods	Canonical successes	Concrete anomalies or unresolved tensions	Key proponents	Representative source
Predictive coding / predictive processing	Cortex implements hierarchical generative models; top-down predictions + bottom-up errors; minimizing prediction error/ free energy organizes perception/ learning	Hierarchical predictive coding models; variational/free-energy formulations; laminar/circuit hypotheses	Explaining extra-classical receptive field effects; unifying story for perception, learning, and some neuroimaging effects	Expectation vs adaptation tensions: repetition suppression can dissociate from expectation manipulation; evidence for “expectation suppression” is inconsistent across contexts; falsification criteria debated	Karl Friston ³² ; Rajesh Rao ³³ ; Georg Keller ³⁴	³⁵
Deep learning / neural networks	Task-optimized deep nets can model sensory and cognitive computations; predictive accuracy can substitute for strong mechanistic interpretability	Training deep nets; encoding/ decoding models; representational similarity; benchmarks	Ventral-stream predictivity from performance-optimized models; “neural predictivity” benchmarking	Behavioral divergences: texture bias vs human shape bias; adversarial vulnerability; critiques that “neural predictivity” may not imply explanatory adequacy; debate over whether current DNNs capture human-like vision	Yann LeCun ³⁶ ; Geoffrey Hinton ³⁷ ; Nikolaus Kriegeskorte ³⁸ ; James DiCarlo ³⁹	⁴⁰

Paradigm	Core assumptions	Typical methods	Canonical successes	Concrete anomalies or unresolved tensions	Key proponents	Representative source
Reinforcement learning models	Behavior best analyzed as reward-maximizing learning; dopamine and basal ganglia implement/ approximate TD-like errors or policy/value updates	MDP/POMDP models; TD learning; model-based vs model-free comparisons; behavioral model fitting	Dopamine response patterns consistent with reward prediction error; formal RL frameworks for learning	Dopamine heterogeneity: dopamine encodes sensory/ motor/cognitive variables; movement-related dopamine and salience signals challenge a simple scalar RPE story; ongoing refinements to reconcile	Richard Sutton ⁴¹ ; Andrew Barto ⁴² ; Wolfram Schultz ⁴³	⁴⁴
Dynamical systems / attractors	Computation is state evolution; low-dimensional trajectories; stable attractors can implement memory/ decisions; dynamics-first explanations can replace “coding” metaphors	Dimensionality reduction; dynamical systems analysis; RNN modeling; attractor theory	Dynamical-systems framing of motor cortex; attractor-based decision circuits	Working memory tensions: evidence for “activity-silent” states and reactivation challenges strict persistent-activity attractor claims; debates over whether dynamics explain “representation”	Krishna Shenoy ⁴⁵ ; David Sussillo ⁴⁶ ; Mark Stokes ⁴⁷	⁴⁸
Connectomics	Structure (wiring diagrams) constrains function; mapping connectivity at scale will enable mechanistic theories	EM reconstruction; diffusion MRI; graph/network analyses; cell-type atlases	C. elegans connectome; large partial insect brain connectomes; human macro-connectome projects; connectome concept framing	Understanding gap: detailed maps do not automatically yield algorithmic/ functional understanding; critique shows standard analysis may fail even with full ground truth in engineered systems	Olaf Sporns ⁴⁹ ; Eric Jonas ⁵⁰ ; Konrad Kording ⁵¹	⁵²

Paradigm dossiers

This section treats each paradigm as a Kuhnian “disciplinary matrix”: core commitments, normal-science puzzle sets, and anomalies that either persist or force boundary-setting.

Single-neuron biophysics

Core assumptions. Neurons are physical devices whose computations are constrained by membrane conductances, morphology, synaptic biophysics, and ionic reversal potentials; mechanistic models can explain and predict spike generation and subthreshold integration. The Hodgkin–Huxley paper is a canonical exemplar, explicitly aiming for equations “sufficiently simple” for theoretical calculation while grounding them in conductance changes. ⁵³

Methods. Conductance-based ODE models and multi-compartment PDE/ODE models, typically fit to voltage-clamp/current-clamp data; simulation environments such as NEURON are designed to make these models practical and linked to data. ⁵⁴

Typical problem sets. Predict action potential shapes and refractory periods; explain firing patterns (bursting, adaptation); map channel expression to excitability; connect dendritic/synaptic mechanisms to input–output transformations; scale from single cells to microcircuits (e.g., “digital reconstructions”). ⁵⁵

Concrete anomalies and tensions.

Parameter degeneracy and identifiability are persistent tensions. Empirically, identified neurons can show large variability in channel expression across individual animals while maintaining function, challenging the inference that “the” correct parameter set is recoverable from limited recordings. ⁵⁶ Mechanistically similar circuit activity can arise from disparate parameters, as shown in the classic demonstration that many circuit parameterizations reproduce similar rhythmic network output. ⁵⁷ Methodologically, “sloppy” parameter sensitivities—where predictions are insensitive along many parameter directions—formalize why fitting detailed models can fail to uniquely identify mechanism even when the model class is correct. ⁵⁸

Scaling detailed reconstructions from sparse data is another tension. Large reconstructions can be scientifically valuable, but the strategy of algorithmically filling gaps (from incomplete physiology/anatomy) makes validation and uniqueness challenging, encouraging debates about what counts as explanatory success: a plausible reconstruction, or a uniquely supported one. ⁵⁹

Competing paradigms. Rate- and spike-based network models often treat single-neuron biophysics as “implementation detail,” while deep learning models sometimes bypass neuron-level realism entirely in favor of predictive accuracy. These are not always direct competitors (Marr-level mismatch), but they compete for explanatory attention and funding/benchmarks in practice. ⁶⁰

Rate-based networks

Core assumptions. Many circuit computations operate on time scales and population aggregates where firing rate is an adequate state variable; spikes can be coarse-grained without losing the phenomenon of

interest. Wilson–Cowan equations are the canonical exemplar: coupled nonlinear population dynamics for excitatory and inhibitory pools. ⁶¹

Methods. Nonlinear dynamical systems (ODE/PDE), stability analysis, mean-field approximations, bifurcation/phase-plane approaches; sometimes linked to data via population averaging or latent-variable fits. ⁶²

Typical problem sets. Explain oscillations, gain control, working memory (persistent activity), decision dynamics, and coarse circuit motifs (E–I balance) in a tractable mathematical form. ⁶³

Concrete anomalies and tensions.

Millisecond spike timing can be reproducible under realistic fluctuating inputs, challenging the idea that only rates matter for cortical computation; this is a direct counterinstance to the strongest “rate-only” interpretations. ⁶⁴

Correlations and population structure can contribute materially to information, beyond what is captured by independent-rate descriptions. A well-known example analyzes a “complete” population of retinal ganglion cells and shows that capturing detailed spatiotemporal correlations changes decoding/information estimates relative to simplified assumptions. ⁶⁵

Irregular firing statistics can conflict with naive temporal integration pictures. The classic argument that highly irregular cortical firing is inconsistent with temporal integration of random EPSPs (under simple assumptions) is often treated as an impetus for more sophisticated network states (e.g., balanced excitation/inhibition) and/or spike-based theories. ⁶⁶

Competing paradigms. Spiking-network frameworks compete most directly (they can reduce to rate models under assumptions, but dispute when those assumptions hold). Coding/metaphor critiques and dynamical-systems approaches also challenge whether “rates encoded in neurons” is the right explanatory primitive. ⁶⁷

Spiking networks

Core assumptions. Spikes are meaningful discrete events; the timing and correlation structure of spikes is central to neural computation; synaptic plasticity rules operating on spikes can implement learning. Spiking neuron modeling textbooks frame spike timing, population codes, and plasticity as core objects of theory. ⁶⁸

Methods. Point-neuron models (integrate-and-fire variants), spike-train statistics, point-process/GLM models, synaptic plasticity models (including STDP), network simulations at scale, and decoding-based evaluation (predicting stimuli/choices from spikes). ⁶⁹

Typical problem sets. Receptive field estimation, neural coding and decoding, oscillations and synchrony, spike-based learning rules, mechanistic models of decision circuits, and spike timing vs rate comparisons. ⁷⁰

Concrete anomalies and tensions.

The field contains an internal empirical tension: some experiments emphasize reliable millisecond spike timing under fluctuating inputs (supporting temporal coding), while other influential work emphasizes high trial-to-trial variability and argues for a probabilistic/rate interpretation in many cortical contexts. ⁷¹ This is not merely “noise”: it changes what counts as the relevant explanatory primitive (exact spike times vs distributional summaries), a Kuhnian-style shift in what puzzles are legitimate. ⁷²

The “neural coding” concept itself is contested. A prominent critique argues that “coding” often depends on experimental context and can mischaracterize the circular causal structure of brains; published responses contest that critique and defend coding-style decomposition as coherent and indispensable. This is a Kuhn-relevant dispute because it targets the paradigm’s conceptual foundations (what a spike train *is* evidence of). ⁷³

Competing paradigms. Rate-based models provide a simplifying rival; dynamical-systems approaches can treat spikes as one realization of latent dynamics; Bayesian population-coding frameworks reinterpret spiking variability as representational rather than merely noisy. ⁷⁴

Bayesian brain and probabilistic coding

Core assumptions. Perception and action can be modeled as inference under uncertainty; the brain represents probability distributions (priors/likelihoods/posteriors) or functionally equivalent quantities; behavior may approximate Bayesian optimality under constraints. A canonical review explicitly coins a “Bayesian coding hypothesis” and highlights uncertainty representation as central. ⁷⁵

Methods. Ideal observer models, Bayesian model comparison, probabilistic population codes (PPCs), and psychophysical experiments manipulating uncertainty to test predicted integration rules. ⁷⁶

Typical problem sets. Cue integration, perceptual inference, sensorimotor learning, confidence/decision-making, and normative explanations of biases as rational under priors and noise. ⁷⁷

Concrete anomalies and tensions.

Empirical suboptimality is a sustained tension. A major synthesis argues that suboptimalities are extensive and that broad “optimality” claims risk becoming weakly informative unless paired with explicit constraints and falsifiable alternatives. ⁷⁸

Model flexibility is a methodological anomaly in Kuhnian terms: critics argue Bayesian models can be tuned through priors and likelihood choices in ways that risk post hoc rationalization (“just-so stories”), undermining their ability to decisively arbitrate between theories. ⁷⁹ Notably, early Bayesian-brain framing itself acknowledged the thinness of neurophysiological tests for uncertainty representation at the time, effectively labeling a core empirical gap. ⁸⁰

Some perceptual phenomena motivate “anti-Bayesian” labels under naive formulations, driving hybrid theories that combine Bayesian decoding with efficient coding constraints. This illustrates Kuhn’s “puzzle-solving” by extending the disciplinary matrix (adding constraints) rather than abandoning it. ⁸¹

Competing paradigms. Predictive coding claims to provide a mechanistic route to approximate Bayesian inference (generative models and prediction errors). Dynamical-systems approaches often reject the need for explicit probabilistic representation, focusing on trajectories and task dynamics instead. ⁸²

Predictive coding and predictive processing

Core assumptions. Hierarchical generative models explain perception: higher areas generate predictions of lower-level activity; feedforward pathways transmit prediction errors; learning updates synapses to reduce expected error. This is explicit in classic predictive coding formulations and in free-energy-based accounts that treat perception and learning as variational inference. ⁸³

Methods. Hierarchical models for sensory signals; variational/free-energy derivations; experimental paradigms manipulating expectations and measuring “prediction error” signatures in spiking or BOLD/EEG/MEG; circuit-level hypotheses about laminar segregation of predictions and errors. ⁸⁴

Typical problem sets. Extra-classical receptive field effects, mismatch responses, contextual modulation, perceptual inference, learning, and unifying accounts across perception and action. ⁸⁵

Concrete anomalies and tensions.

A recurring empirical tension is the relationship between repetition suppression, adaptation, and expectation/prediction. In macaque inferior temporal cortex, manipulation of repetition probability did not affect repetition suppression in a protocol designed to test expectation-based accounts, suggesting that suppression can be dominated by bottom-up adaptation mechanisms rather than top-down expectation.

⁸⁶ A broad review of expectation suppression evidence argues that results are inconsistent in many contexts and that null results are informative for theory design, directly challenging simplistic “expectation always suppresses responses” narratives. ⁸⁷

Falsifiability remains an explicit concern even within supportive treatments: major perspectives discuss how predictive processing might be implemented at circuit level and how it *could* be falsified, implying that many current uses remain underconstrained. ⁸⁸

Competing paradigms. Deep learning provides a competing explanation for sensory hierarchies: task-optimized feedforward (and recurrent) networks can match neural responses without explicit prediction-error architectures. Bayesian brain formulations can subsume predictive coding as one mechanistic route, but also compete with it over what is essential (explicit prediction errors vs probabilistic computations more broadly). ⁸⁹

Deep learning and neural networks

Core assumptions. Neural systems can be modeled as layered networks whose representations are shaped by optimization for tasks; matching brain data may be achieved by matching task performance and then comparing internal representations to neural measurements. This view is prominent in deep-learning reviews and in neuroscience-focused treatments arguing DNNs provide a new modeling framework for biological vision. ⁹⁰

Methods. Train deep nets (often on labeled or self-supervised tasks), then evaluate predictivity against neural recordings via encoding models/representational similarity; use benchmarks and platforms to compare many models systematically. ⁹¹

Typical problem sets. Modeling ventral visual stream; building neurally predictive models of sensory cortex; extending toward language and higher cognition through benchmark-driven model selection. ⁹²

Concrete anomalies and tensions.

Systematic behavioral divergences between standard CNNs and humans are well documented. For example, ImageNet-trained CNNs can be strongly biased toward texture cues in ways that conflict with human shape-biased recognition, demonstrating that high task accuracy need not imply human-like strategies. ⁹³

Adversarial vulnerability is a second foundational anomaly: small, structured perturbations can induce confident misclassification, revealing fragilities not characteristic of human perception and raising questions about whether current network families capture the invariances used by biological systems. ⁹⁴

At the level of scientific standards, a prominent debate argues that “neural predictivity” approaches (including benchmark-driven selection) risk conflating partial correlations with explanatory understanding, and that current DNNs face “deep problems” as models of human vision. This represents a Kuhnian-style foundational dispute about what should count as success. ⁹⁵

Competing paradigms. Predictive coding offers a generative alternative emphasizing top-down prediction errors; dynamical-systems/RNN approaches emphasize recurrent dynamics and low-dimensional trajectories rather than feedforward feature hierarchies; biophysical models challenge the biological plausibility of backprop-like learning in cortex, motivating “neuro-inspired” learning alternatives and integration proposals. ⁹⁶

Reinforcement learning models

Core assumptions. Learning is driven by reward-based prediction errors; behavior can be modeled as policy/value learning in MDP-like environments; neural substrates (especially dopamine/basal ganglia loops) implement learning signals analogous to temporal-difference errors or related quantities. ⁹⁷

Methods. Behavioral task modeling with RL variants (model-free, model-based, distributional, hierarchical RL); fitting trial-by-trial learning curves; measuring dopamine and striatal signals; causal manipulations. ⁹⁸

Typical problem sets. Pavlovian and instrumental learning, exploration/exploitation, dopamine-related learning, computational psychiatry constructs, and mapping algorithmic RL quantities to neural data. ⁹⁹

Concrete anomalies and tensions.

A major tension is dopamine heterogeneity: dopamine neurons can encode non-reward variables (salience, sensory/motor/cognitive variables), undermining a one-dimensional “RPE-only” interpretation. A major

review proposes multiple dopamine neuron types with distinct roles, explicitly broadening the functional space beyond reward value. ¹⁰⁰

More recent large-scale recordings show dopamine neurons encoding arrays of sensory, motor, and cognitive variables in decision tasks, directly demonstrating mixed selectivity inconsistent with a single scalar teaching signal. ¹⁰¹

Explicit counter-claims argue that dopamine signals in some pathways are better understood as movement-related rather than reward-prediction signals (“reward prediction error does not explain movement selectivity”), and even recent high-profile work points to movement encoding as central in certain tasks. ¹⁰²

From a Kuhnian perspective, these are classic “anomalies” that have driven (i) proliferation of alternatives (value vs salience vs movement accounts) and (ii) patching/expansion of the paradigm (multi-factor dopamine theories, richer RL variants) rather than a wholesale abandonment of RL. ¹⁰³

Competing paradigms. Predictive processing and Bayesian control accounts can reinterpret learning as minimizing surprise or performing inference in latent-state models, potentially reframing what dopamine is “for.” Dynamical-systems views can emphasize policy execution dynamics and closed-loop control rather than error coding. ¹⁰⁴

Dynamical systems and attractor networks

Core assumptions. Neural computation is well described as the evolution of population state in time; low-dimensional manifolds and dynamics may be more explanatory than single-neuron tuning curves; attractors (fixed points, line/ring attractors) can implement memory, decisions, and stable representations. A dynamical-systems perspective on motor cortex is explicitly framed as a unifying conceptual approach. ¹⁰⁵

Methods. State-space modeling, dimensionality reduction, dynamical systems identification, recurrent neural networks as phenomenological models, and attractor analyses; linking dynamics to behavior (e.g., movement generation). ¹⁰⁶

Typical problem sets. Motor control trajectories, preparatory dynamics, decision integration dynamics, working memory maintenance, and linking population activity patterns to computation. ¹⁰⁷

Concrete anomalies and tensions.

Working memory is a focal site of tension. Classical attractor accounts emphasize persistent activity states; however, “activity-silent” working memory frameworks argue that information can be maintained in hidden/synaptic states with dynamic population activity, and that active representations can drop to baseline and then be reactivated. ¹⁰⁸ This is a direct counterinstance to the strongest form of persistent-activity-as-necessary, forcing either boundary-setting (persistent activity in some regimes) or paradigm modification (hybrid attractor + silent states). ¹⁰⁹

Conceptually, dynamical-systems approaches can conflict with “neural coding” interpretations over what counts as representation and explanation, a dispute that is actively discussed in the field’s conceptual literature. ¹¹⁰

Competing paradigms. Predictive coding and Bayesian frameworks emphasize inference and explicit uncertainty; deep learning emphasizes task-optimized representational hierarchies; dynamical systems can coexist with both but can also substitute for “coding” explanations in some domains (notably motor cortex). ¹¹¹

Connectomics

Core assumptions. Mapping connectivity is foundational: the connectome is a structural substrate from which function and dysfunction can be inferred; comprehensive maps (synaptic or macro-scale) will enable mechanistic understanding of circuits and cognition. The “connectome” framing explicitly positions connection matrices as indispensable and proposes strategies for achieving them. ¹¹²

Methods. Dense EM reconstruction; diffusion MRI and tractography for macro-connectomes; graph/network science analyses; cell-type atlases and cross-connectome comparisons; large collaborative datasets. ¹¹³

Typical problem sets. Produce wiring diagrams at multiple scales; identify motifs/modules/hubs; relate structural to functional connectivity; constrain dynamic models with anatomical priors. ¹¹⁴

Concrete anomalies and tensions.

A major anomaly is the “understanding gap”: even with extremely rich structural and activity data, standard analysis approaches may fail to yield a satisfying mechanistic explanation. The microprocessor “model organism” critique shows that applying common neuroscience analysis tools to a fully understood engineered system produces many familiar-looking results but not an actual understanding of information processing in the engineering sense. This challenges the implicit inference “more data + current methods → understanding,” a core motivation sometimes associated with large-scale mapping programs. ¹¹⁵

A second tension is that structure alone is insufficient: synaptic weights, neuromodulation, state dependence, and plasticity shape function. This is not a refutation of connectomics but a boundary condition that forces integration with physiological and computational paradigms. The existence of work attempting to predict functional connectivity from the connectome illustrates both the promise and the difficulty of bridging structure to dynamics. ¹¹⁶

Competing paradigms. Dynamical systems and spiking/rate network modeling compete over what is explanatory-primitive (wiring vs state evolution vs computation). Kuhn would predict that without shared standards for “understanding,” connectomics debates will persist as paradigm boundary disputes rather than decisive falsifications. ¹¹⁷

Interparadigm conflicts and debates

This section identifies paradigm clashes where commitments are not merely complementary (Marr-level separation) but genuinely competing in Kuhn's sense: conflicting standards about what counts as explanation and what anomalies matter.

Rate versus timing in neural codes

A long-running debate centers on whether spike *counts/rates* or precise *timing* is the primary explanatory currency. Evidence for precise timing reliability under fluctuating inputs supports timing-centric views, whereas evidence emphasizing variability and probabilistic interpretations supports rate/distribution-centric views. ⁷¹

The dispute is Kuhnian because it shifts exemplar problems: rate-centric paradigms treat irregular firing as a feature (stochastic computation, distributional codes), while timing-centric paradigms treat millisecond precision as evidence for temporally structured computation. The literature explicitly positions these as implications for “information coding,” i.e., conceptual foundations of the field. ¹¹⁸

Bayesian optimality versus suboptimality and constraint-based accounts

Bayesian-brain approaches often frame behavior as “near-optimal” under uncertainty, supported by canonical cue integration results. ¹¹⁹ Critics argue that pervasive suboptimality, plus the flexibility of priors/likelihoods, can make “optimality” claims difficult to falsify, turning them into weak constraints on theory choice. ¹²⁰ A key contemporary move is to shift the debate from “optimal vs not” to “optimal under which biologically grounded constraints,” which is closer to a Lakatosian repair strategy but also resembles Kuhnian puzzle-expansion during periods of contested foundations. ¹²¹

Predictive coding versus adaptation and alternative explanations

Predictive processing predicts systematic roles for expectations and prediction errors. Yet empirical work shows contexts where repetition suppression appears insensitive to expectation manipulations, implying that some signatures can arise from adaptation mechanisms without predictive coding. ¹²² Recent broad evaluations of expectation suppression emphasize inconsistent evidence across contexts, reinforcing that experimental design and confounds strongly matter. ⁸⁷

These tensions are paradigmatic because they contest what counts as a “prediction error signature” and whether the program has a distinctive empirical fingerprint or can always be accommodated by auxiliary hypotheses. Kuhn treats such underdetermination as a hallmark of pre-crisis stress and interparadigm argument. ⁶

Deep neural networks as brain models versus mechanistic/normative demands

Task-trained deep networks can predict neural responses (a major success), and benchmarking platforms formalize this predictive standard. ⁹¹ However, critics argue that predictive correlations do not guarantee mechanistic insight or human-like computation, and that DNNs show systematic behavioral failures (texture bias, adversarial vulnerability) that are not minor discrepancies but structural mismatches. ¹²³

This is a direct Kuhnian conflict over standards: “predictive accuracy on neural data” (a benchmark culture) versus “explanatory adequacy/human-like computation/robustness” (a cognitive/psychophysical standard). As Kuhn would expect, disagreement persists partly because success criteria differ. ¹²⁴

Dopamine and RL teaching signals versus multi-factor neuromodulatory roles

The RPE account is a canonical exemplar linking dopamine responses to TD-like errors. ¹²⁵ Yet multiple lines of evidence support broader encoding (value and salience types; multidimensional sensory/motor/cognitive variables; movement selectivity), leading to pluralistic reinterpretations rather than a single clean mapping. ¹²⁶

This represents interparadigm pressure between a clean algorithmic mapping (scalar error) and a richer biological reality (heterogeneous signals), with active debate about whether to (i) expand RL formalisms, (ii) reinterpret dopamine’s computational role, or (iii) treat dopamine as one of multiple learning/control signals. ¹²⁷

Structure-first connectomics versus dynamics-first explanations

Connectomics posits structural maps as indispensable foundations; dynamical systems approaches emphasize that function is realized in state trajectories and context-dependent dynamics. ¹²⁸ The microprocessor critique sharply challenges the inference that current methods applied to complete structural/activity data suffice for mechanistic understanding, providing a widely cited cautionary example. ¹¹⁵

This conflict is Kuhnian in that it targets “what counts as understanding”: a wiring diagram, or a functional decomposition into algorithms and mechanisms. In practice, modern work increasingly treats connectomes as constraints for dynamical models, suggesting synthesis rather than replacement. ¹²⁹

Crisis assessment and timelines

Timeline of paradigm emergence and stress points

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timeline
  title Computational neuroscience paradigms and debated anomalies
  1952 : Hodgkin-Huxley conductance model becomes a core exemplar
  1972 : Wilson-Cowan population dynamics formalize rate-based circuits
  1982 : Hopfield networks popularize attractors as computational primitives
  1993-1998 : Spike variability vs timing reliability debates sharpen
  (irregularity; variability; reliability)
  1999 : Predictive coding formalized for visual cortex hierarchies
  2004-2006 : Bayesian brain and probabilistic population codes consolidate
  2005 : "Connectome" framed as a foundational structural target
  2009-2013 : Dynamical systems perspective crystallizes in motor control
  reviews
  2014-2016 : Task-trained deep nets predict ventral-stream responses;
  integration calls emerge
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2017 : Microprocessor critique raises alarms about method-to-understanding gap
 2018-2021 : Predictive processing and expectation suppression evidence debated; suboptimality reviews consolidate
 2023 : High-profile critiques of DNNs as human vision models intensify
 2018-2025 : Dopamine signals broaden beyond scalar RPE in major studies

Key anchors for this timeline include Hodgkin–Huxley (1952), Wilson–Cowan (1972), Hopfield (1982), spike variability and timing debates (1993–1998), predictive coding (1999), Bayesian brain and PPCs (2004–2006), connectome framing (2005), dynamical systems motor control framing (2011–2013), deep nets predicting ventral stream (2014), method-to-understanding critique (2017), debates over expectation suppression and optimality/suboptimality (2018–2021), and dopamine/RL reinterpretations (2018 onward). ¹³⁰

Are there signs of Kuhnian crisis

A strict Kuhnian crisis presupposes a *dominant paradigm* whose puzzle-solving capacity is doubted, producing a community-wide crisis and then replacement. On that standard, **computational neuroscience as a whole does not appear to be in a single crisis**, because it does not operate under one dominant paradigm with shared success criteria. Instead, it more closely resembles a stable (if contentious) multi-paradigm ecology. The very existence of integrative calls framing the landscape as multiple disciplines requiring assembly is evidence against a single “normal science” regime. ¹³¹

That said, several *localized Kuhnian crisis signals* are visible:

Persistent anomalies that strike near core commitments.

- In biophysical modeling, degeneracy/identifiability issues do not merely add noise; they challenge whether detailed parameterized models can deliver unique mechanistic explanations from realistic datasets. The persistence of this theme across decades (channel variability in 2006; “sloppiness” formalism; circuit degeneracy demonstrations) indicates a stable anomaly cluster. ¹³²
- In predictive processing, expectation-based predictions face repeated empirical complications (e.g., expectation manipulations failing to modulate repetition suppression in some paradigms; inconsistent evidence for expectation suppression across contexts), forcing either major boundary-setting or increasingly elaborate auxiliary hypotheses. ¹³³
- In deep learning as cognitive/neural theory, systematic mismatches (robustness, texture bias) plus high-profile critiques of explanatory adequacy suggest that “task accuracy + neural predictivity” may be insufficient as a dominant standard. ¹²³
- In RL neuroscience, dopamine heterogeneity and movement/salience signals are persistent and expanding, challenging a clean mapping from dopamine to scalar RPE, though the RL program often absorbs this by extending the model class. ¹²⁶

Loss of confidence expressed in major venues.

- The Bayesian optimality program has faced explicit “just-so story” critiques and major reviews emphasizing that broad optimality claims can be “virtually meaningless” without stronger constraints, which is a Kuhn-relevant signal of stress about the paradigm’s explanatory standards. ⁷⁹

- The DNN-as-brain-model program has faced major target-article critiques in a top debate venue, indicating substantial community disagreement about the paradigm's adequacy rather than routine puzzle-solving. ¹³⁴
- The connectomics/big-data promise is explicitly problematized by a widely cited critique demonstrating the failure of common analysis pipelines to yield understanding even with full ground truth, undermining a key optimism narrative. ¹³⁵

Proliferation of alternatives and hybridization.

Rather than convergence onto a single successor paradigm, the field exhibits proliferation plus synthesis: integrative benchmarking platforms, calls to integrate neuroscience with deep learning, and frameworks like “cognitive computational neuroscience” aim to combine modeling cultures rather than replace one with another. This pattern is more consistent with field expansion and methodological consolidation than with Kuhnian revolution. ¹³⁶

What “crisis” looks like here, if not a revolution

From a Kuhnian perspective, the most plausible diagnosis is:

- Computational neuroscience is *not fully in a normal-science phase* at the field level: it contains multiple partly incommensurable paradigms and standards. ¹³⁷
- The field exhibits *recurring local crises* at interfaces where explanatory standards collide: normative optimality versus biological constraints; predictive-processing signatures versus adaptation/confounds; predictive accuracy benchmarks versus explanatory adequacy; clean algorithmic mappings versus biological heterogeneity. ¹³⁸
- Resolution tends toward *hybrid disciplinary matrices* (benchmarking + mechanistic constraints; Bayesian decoding + efficient coding; connectome-informed dynamical models) rather than abrupt replacement. ¹³⁹

In Kuhnian terms, this resembles an extended period in which multiple paradigms coexist because none can credibly absorb the full anomaly set across levels—and because “the rules of the game” differ by level and data modality. ¹⁴⁰

Synthesis and Kuhnian interpretation

A Kuhnian reading suggests that computational neuroscience's most important “paradigm conflicts” are not merely technical; they are conflicts over **explanatory aims** and **evaluation standards**:

- Mechanistic paradigms (biophysics, spiking circuits) often treat interpretability and causal mapping as non-negotiable, but face degeneracy/identifiability anomalies that complicate claims of uniquely correct mechanism. ¹⁴¹
- Normative paradigms (Bayesian brain, RL) often treat optimality/inference as organizing principles, but face persistent empirical suboptimalities and mapping ambiguities (priors, dopamine heterogeneity). ¹⁴²
- Data-driven paradigms (deep learning, connectomics) often treat predictive performance and scale as organizing principles, but face anomalies about robustness and the gap between prediction and explanation. ¹⁴³

- Dynamical systems approaches function as both competitor and mediator, reframing coding debates and offering shared mathematical language that can integrate mechanistic constraints with behavioral targets, but they also collide with representational/coding intuitions. ¹⁴⁴

If one insists on a single Kuhnian verdict: computational neuroscience today looks less like a field awaiting one revolution and more like a field whose subject matter (brains as multiscale adaptive systems) makes **stable pluralism** a rational equilibrium. In that equilibrium, “crisis” is episodic and interface-specific, and progress often comes from *redefining puzzles* (benchmarks, constraints, and levels of explanation) rather than overthrowing a single paradigm. ¹⁴⁵

¹ ⁶ ⁷ ¹¹ ²³ ²⁵ ²⁹ ³⁴ ⁴¹ ⁷² ¹¹⁷ ¹²⁴ ¹³⁷ ¹⁴⁰ ¹⁴⁵ <https://www.lri.fr/~mbl/Stanford/CS477/papers/Kuhn-SSR-2ndEd.pdf>

<https://www.lri.fr/~mbl/Stanford/CS477/papers/Kuhn-SSR-2ndEd.pdf>

² ¹⁰ ¹⁴ ¹⁵ ¹⁹ ³⁷ ⁴⁶ ⁵³ ⁵⁵ ¹³⁰ <https://pubmed.ncbi.nlm.nih.gov/12991237/>

<https://pubmed.ncbi.nlm.nih.gov/12991237/>

³ ¹⁴¹ <https://pubmed.ncbi.nlm.nih.gov/16791145/>

<https://pubmed.ncbi.nlm.nih.gov/16791145/>

⁴ ⁷⁸ ¹²⁰ ¹²¹ ¹³⁸ ¹⁴² <https://pmc.ncbi.nlm.nih.gov/articles/PMC6110994/>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC6110994/>

⁵ ⁹ ¹² <https://plato.stanford.edu/entries/thomas-kuhn/>

<https://plato.stanford.edu/entries/thomas-kuhn/>

⁸ ⁶⁰ <https://people.ciirc.cvut.cz/~hlavac/pub/MiscTextForStudents/1982MarrDavidVisionBook.pdf>

<https://people.ciirc.cvut.cz/~hlavac/pub/MiscTextForStudents/1982MarrDavidVisionBook.pdf>

¹³ ³³ ⁴⁵ ⁷⁷ ¹¹⁹ <https://www.nature.com/articles/nature02169>

<https://www.nature.com/articles/nature02169>

¹⁶ ⁹³ ¹²³ <https://arxiv.org/abs/1811.12231>

<https://arxiv.org/abs/1811.12231>

¹⁷ ⁴⁰ ⁹⁰ <https://www.nature.com/articles/nature14539>

<https://www.nature.com/articles/nature14539>

¹⁸ ⁵⁶ ¹³² <https://pubmed.ncbi.nlm.nih.gov/16444270/>

<https://pubmed.ncbi.nlm.nih.gov/16444270/>

²⁰ ²⁴ ³⁹ ⁸⁹ ⁹¹ ⁹² <https://www.pnas.org/doi/10.1073/pnas.1403112111>

<https://www.pnas.org/doi/10.1073/pnas.1403112111>

²¹ ⁹⁵ ¹³⁴ ¹⁴³ <https://www.cambridge.org/core/journals/behavioral-and-brain-sciences/article/deep-problems-with-neural-network-models-of-human-vision/ABCE483EE95E80315058BB262DCA26A9>

<https://www.cambridge.org/core/journals/behavioral-and-brain-sciences/article/deep-problems-with-neural-network-models-of-human-vision/ABCE483EE95E80315058BB262DCA26A9>

²² ³⁸ ⁶¹ ⁶² ⁶³ <https://pmc.ncbi.nlm.nih.gov/articles/PMC1484078/>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC1484078/>

²⁶ ⁶⁸ ⁶⁹ <https://lcnwww.epfl.ch/gerstner/PUBLICATIONS/SpikingNeuronM-extracts.pdf>

<https://lcnwww.epfl.ch/gerstner/PUBLICATIONS/SpikingNeuronM-extracts.pdf>

27 116 129 <https://www.sciencedirect.com/science/article/pii/S0960982221003432>
<https://www.sciencedirect.com/science/article/pii/S0960982221003432>

28 108 <https://pubmed.ncbi.nlm.nih.gov/26051384/>
<https://pubmed.ncbi.nlm.nih.gov/26051384/>

30 82 84 104 <https://pmc.ncbi.nlm.nih.gov/articles/PMC1569488/>
<https://pmc.ncbi.nlm.nih.gov/articles/PMC1569488/>

31 36 42 75 80 <https://pubmed.ncbi.nlm.nih.gov/15541511/>
<https://pubmed.ncbi.nlm.nih.gov/15541511/>

32 100 103 126 127 <https://pubmed.ncbi.nlm.nih.gov/21144997/>
<https://pubmed.ncbi.nlm.nih.gov/21144997/>

35 43 47 83 85 96 https://www.nature.com/articles/nn0199_79
https://www.nature.com/articles/nn0199_79

44 97 98 <https://web.stanford.edu/class/psych209/Readings/SuttonBartoIPRLBook2ndEd.pdf>
<https://web.stanford.edu/class/psych209/Readings/SuttonBartoIPRLBook2ndEd.pdf>

48 105 106 107 111 144 <https://pubmed.ncbi.nlm.nih.gov/23725001/>
<https://pubmed.ncbi.nlm.nih.gov/23725001/>

49 64 70 71 <https://pubmed.ncbi.nlm.nih.gov/7770778/>
<https://pubmed.ncbi.nlm.nih.gov/7770778/>

50 74 76 <https://pubmed.ncbi.nlm.nih.gov/17057707/>
<https://pubmed.ncbi.nlm.nih.gov/17057707/>

51 79 <https://pubmed.ncbi.nlm.nih.gov/22545686/>
<https://pubmed.ncbi.nlm.nih.gov/22545686/>

52 112 128 <https://pmc.ncbi.nlm.nih.gov/articles/PMC1239902/>
<https://pmc.ncbi.nlm.nih.gov/articles/PMC1239902/>

54 <https://pubmed.ncbi.nlm.nih.gov/9248061/>
<https://pubmed.ncbi.nlm.nih.gov/9248061/>

57 <https://www.pnas.org/doi/abs/10.1073/pnas.2207632119>
<https://www.pnas.org/doi/abs/10.1073/pnas.2207632119>

58 <https://journals.plos.org/ploscompbiol/article?id=10.1371%2Fjournal.pcbi.0030189>
<https://journals.plos.org/ploscompbiol/article?id=10.1371%2Fjournal.pcbi.0030189>

59 <https://www.sciencedirect.com/science/article/pii/S0092867415011915>
<https://www.sciencedirect.com/science/article/pii/S0092867415011915>

65 <https://pmc.ncbi.nlm.nih.gov/articles/PMC2684455/>
<https://pmc.ncbi.nlm.nih.gov/articles/PMC2684455/>

66 <https://pubmed.ncbi.nlm.nih.gov/8423479/>
<https://pubmed.ncbi.nlm.nih.gov/8423479/>

67 118 <https://www.jneurosci.org/content/18/10/3870>
<https://www.jneurosci.org/content/18/10/3870>

- 73 110 <https://www.cambridge.org/core/journals/behavioral-and-brain-sciences/article/is-coding-a-relevant-metaphor-for-the-brain/D578626E4888193FFFAE5B6E2C37E052>
<https://www.cambridge.org/core/journals/behavioral-and-brain-sciences/article/is-coding-a-relevant-metaphor-for-the-brain/D578626E4888193FFFAE5B6E2C37E052>
- 81 <https://pubmed.ncbi.nlm.nih.gov/26343249/>
<https://pubmed.ncbi.nlm.nih.gov/26343249/>
- 86 122 133 <https://academic.oup.com/cercor/article/21/7/1547/333836>
<https://academic.oup.com/cercor/article/21/7/1547/333836>
- 87 <https://www.sciencedirect.com/science/article/abs/pii/S0149763421001536>
<https://www.sciencedirect.com/science/article/abs/pii/S0149763421001536>
- 88 <https://www.sciencedirect.com/science/article/pii/S0896627318308572>
<https://www.sciencedirect.com/science/article/pii/S0896627318308572>
- 94 <https://arxiv.org/abs/1312.6199>
<https://arxiv.org/abs/1312.6199>
- 99 125 <https://www.science.org/doi/abs/10.1126/science.275.5306.1593>
<https://www.science.org/doi/abs/10.1126/science.275.5306.1593>
- 101 <https://pubmed.ncbi.nlm.nih.gov/31142844/>
<https://pubmed.ncbi.nlm.nih.gov/31142844/>
- 102 <https://elifesciences.org/articles/42992>
<https://elifesciences.org/articles/42992>
- 109 <https://www.sciencedirect.com/science/article/pii/S1364661315001023>
<https://www.sciencedirect.com/science/article/pii/S1364661315001023>
- 113 114 <https://pmc.ncbi.nlm.nih.gov/articles/PMC3724347/>
<https://pmc.ncbi.nlm.nih.gov/articles/PMC3724347/>
- 115 135 <https://pmc.ncbi.nlm.nih.gov/articles/PMC5230747/>
<https://pmc.ncbi.nlm.nih.gov/articles/PMC5230747/>
- 131 <https://pubmed.ncbi.nlm.nih.gov/30127428/>
<https://pubmed.ncbi.nlm.nih.gov/30127428/>
- 136 <https://www.sciencedirect.com/science/article/pii/S089662732030605X>
<https://www.sciencedirect.com/science/article/pii/S089662732030605X>
- 139 <https://pmc.ncbi.nlm.nih.gov/articles/PMC7286693/>
<https://pmc.ncbi.nlm.nih.gov/articles/PMC7286693/>